

UDP

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Chapter 1

udp_lib

Chapter 2

Topic Index

2.1 Topics

Here is a list of all topics with brief descriptions:

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Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

pseudo_header		
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udp_head		
	Struct is header UDP pack	32
udp_pack		
	Struct is UDP package	33

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

udp.c	Code file for work udp package	35
udp.h	Header file for work udp package	38

Chapter 5

Topic Documentation

5.1 work for udp

Group function for work udp package.

Classes

- struct [udp_head](#)
Struct is header UDP pack.
- struct [udp_pack](#)
Struct is UDP package.
- struct [pseudo_header](#)
Struct [pseudo_header](#) for calculate checksum for UDP package.

Macros

- #define [PACKED](#) __attribute__((packed))

Typedefs

- typedef struct [udp_pack](#) * [udp_pack_t](#)
UDP packet descriptor.

Functions

- static uint16_t [inet_port](#) (const char *const port)
Function support for parsing port.
- static void [set_size_udp_pack](#) ([udp_pack_t](#) pack, const uint16_t size)
Function raw write size UDP package.
- static uint32_t [sum_compute](#) (void *ptr, uint16_t nbytes)
Function calculate sum in big endian.
- static uint16_t [checksum_compute](#) (uint32_t sum)
Function calculating from sum big endian to checksum big endian.

- static void [calculate_checksum_udp_pack](#) ([udp_pack_t](#) pack)
Function calculate checksum for ip header and UDP package.
- static void * [get_pack_udp_pack](#) ([udp_pack_t](#) pack)
Function raw get pointer on UDP package.
- static void * [inet_mac](#) (const char *const mac_address)
Function return mac address in big endian.
- [udp_pack_t](#) [init_udp_pack](#) (void)
Function for create object UDP package.
- ssize_t [set_port_source_udp_pack](#) ([udp_pack_t](#) pack, const char *const port)
Function for setting source port in UDP package.
- ssize_t [set_port_destantion_udp_pack](#) ([udp_pack_t](#) pack, const char *const port)
Function for setting destantion port in UDP package.
- ssize_t [set_ip_address_source_udp_pack](#) ([udp_pack_t](#) pack, const char *const ip)
Function for setting source ip address in UDP package.
- ssize_t [set_ip_address_destantion_udp_pack](#) ([udp_pack_t](#) pack, const char *const ip)
Function for setting destantion ip address in UDP package.
- ssize_t [add_data_udp_pack](#) ([udp_pack_t](#) pack, void *data, uint16_t size)
Function addition data in UDP package.
- ssize_t [set_data_udp_pack](#) ([udp_pack_t](#) pack, const void *const data, uint16_t size)
Function overriding data in UDP package.
- ssize_t [add_byte_udp_pack](#) ([udp_pack_t](#) pack, const uint8_t byte)
Function addition byte in UDP package.
- ssize_t [set_input_data_udp_pack](#) ([udp_pack_t](#) pack)
Function read data from stdin in UDP package.
- ssize_t [set_file_data_udp_pack](#) ([udp_pack_t](#) pack, const char *const file_name)
Function read data from file in UDP package.
- ssize_t [set_interface_udp_pack](#) ([udp_pack_t](#) pack, const char *const interface)
Function for pick interface to send UDP package.
- ssize_t [send_udp_pack](#) ([udp_pack_t](#) pack)
Function to send UDP package.
- char * [get_mac_address_source_udp_pack](#) ([udp_pack_t](#) pack)
Function getting mac address for source.
- char * [get_mac_address_destantion_udp_pack](#) ([udp_pack_t](#) pack)
Function setting mac address for destantion.
- ssize_t [set_mac_address_source_udp_pack](#) ([udp_pack_t](#) pack, const char *const mac_address)
Function setting mac address for source.
- ssize_t [set_mac_address_destantion_udp_pack](#) ([udp_pack_t](#) pack, const char *const mac_address)
Function setting mac address for destantion.
- char * [get_interface_udp_pack](#) ([udp_pack_t](#) pack)
Function for getting interface.
- char * [get_port_source_udp_pack](#) ([udp_pack_t](#) pack)
Function for getting port source.
- char * [get_port_destantion_udp_pack](#) ([udp_pack_t](#) pack)
Function for getting port destantion.
- char * [get_ip_address_source_udp_pack](#) ([udp_pack_t](#) pack)
Function for getting ip address source.
- char * [get_ip_address_destantion_udp_pack](#) ([udp_pack_t](#) pack)
Function for getting ip address destantion.
- uint16_t [get_size_data_udp_pack](#) ([udp_pack_t](#) pack)
Function for getting size data.
- char * [get_data_udp_pack](#) ([udp_pack_t](#) pack)

Function for getting data.

- char * [get_data_hex_udp_pack](#) (udp_pack_t pack)

Function for getting data hex.

- ssize_t [print_udp_pack](#) (udp_pack_t pack)

Function print all info about udp pack.

- void [destroy_udp_pack](#) (udp_pack_t pack)

Function free UDP package.

Variables

- struct [udp_head](#) PACKED

5.1.1 Detailed Description

Group function for work udp package.

5.1.2 Macro Definition Documentation

5.1.2.1 PACKED

```
struct pseudo\_header PACKED __attribute__((packed))
```

5.1.3 Typedef Documentation

5.1.3.1 udp_pack_t

```
typedef struct udp\_pack* udp\_pack\_t
```

UDP packet descriptor.

The primary type for working with the library. All functions accept this pointer as an argument.

5.1.4 Function Documentation

5.1.4.1 add_byte_udp_pack()

```
ssize_t add_byte_udp_pack (  
    udp\_pack\_t pack,  
    const uint8_t byte)
```

Function addition byte in UDP package.

Note

You must call `init_udp_pack` before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
in	<i>byte</i>	Addition byte in UDP package.

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
ret = add_byte_udp_pack(pack, 'n');
if (ret)
    goto add_not_byte;
ret = add_byte_udp_pack(pack, '\0');
goto add_not_byte;
add_not_byte:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.2 add_data_udp_pack()

```
ssize_t add_data_udp_pack (
    udp_pack_t pack,
    void * data,
    uint16_t size)
```

Function addition data in UDP package.

Note

You must call `init_udp_pack` before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
in	<i>data</i>	Addition data in UDP package.
in	<i>size</i>	Size addition data for UDP package.

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;
const char * const message = "Hello, world";
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
ret = add_data_udp_pack(pack, message, strlen(message));
if (ret == -1)
    goto add_not_data;
add_not_data:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.3 calculate_checksum_udp_pack()

```
void calculate_checksum_udp_pack (
    udp_pack_t pack) [static]
```

Function calculate checksum for ip header and UDP package.

Parameters

in, out	pack	UDP package for work.
---------	------	-----------------------

Note

This function is private. Not used outside [udp_lib/udp.c](#)

5.1.4.4 checksum_compute()

```
uint16_t checksum_compute (
    uint32_t sum) [static]
```

Function calculating from sum big endian to checksum big endian.

Parameters

in	sum	Sum in big endian.
----	-----	--------------------

Returns

Checksum in big endian.

Note

This function is private. Not used outside [udp_lib/udp.c](#)

5.1.4.5 destroy_udp_pack()

```
void destroy_udp_pack (
    udp_pack_t pack)
```

Function free UDP package.

Note

You must call `init_udp_pack` before this.

Parameters

<code>in, out</code>	<code>pack</code>	UDP package for work. Usage example.
----------------------	-------------------	--------------------------------------

```

ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
// other code whit using udp_pack_t
get_not_udp_pack:
destroy_udp_pack(pack);

```

5.1.4.6 get_data_hex_udp_pack()

```
char * get_data_hex_udp_pack (
    udp_pack_t pack)
```

Function for getting data hex.

Note

You must call `init_udp_pack` before this.

Parameters

<code>in, out</code>	<code>pack</code>	UDP package for work.
----------------------	-------------------	-----------------------

Returns

Data hex in UDP package or NULL pointer on error. Usage example.

```

ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
{
    char * data = get_data_hex_udp_pack(pack);
    if (data == NULL) {
        ret = -1;
        goto get_not_data_hex_udp_pack;
    }
    // other code whit udp_pack_t
}
// other code whit udp_pack_t
get_not_data_hex_udp_pack:
get_not_udp_pack:
destroy_udp_pack(pack);

```

5.1.4.7 get_data_udp_pack()

```
char * get_data_udp_pack (
    udp_pack_t pack)
```

Function for getting data.

Note

You must call [init_udp_pack](#) before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
---------	-------------	-----------------------

Returns

Data in UDP package. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
{
    void * data = get_data_udp_pack(pack);
    // other code whit udp_pack_t
}
// other code whit udp_pack_t
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.8 get_interface_udp_pack()

```
char * get_interface_udp_pack (
    udp_pack_t pack)
```

Function for getting interface.

Note

You must call [init_udp_pack](#) before this.

You can call [set_interface_udp_pack](#) before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
---------	-------------	-----------------------

Returns

String or NULL pointer is fail. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
ret = set_interface_udp_pack(pack, "lo");
if (ret == -1)
    goto setting_not_interface;
{
    char * interface = NULL;
    interface = get_interface_udp_pack(pack);
    if (interface == NULL)
        getting_not_interface;
    // other code whit udp_pack_t
}
// other code whit udp_pack_t
getting_not_interface:
setting_not_interface:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.9 `get_ip_address_destantion_udp_pack()`

```
char * get_ip_address_destantion_udp_pack (
    udp_pack_t pack)
```

Function for getting ip address destantion.

Note

You must call `init_udp_pack` before this.

You can call `set_ip_address_destantion_udp_pack` before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
---------	-------------	-----------------------

Returns

String or NULL pointer is fail. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
ret = set_ip_address_source_udp_pack(pack, "127.0.0.1");
if (ret == -1)
    goto setting_not_ip_destantion;
{
    char * ip_address = NULL;
    ip_address = get_ip_destantion_udp_pack(pack);
    if (ip_address == NULL)
        getting_not_ip_address;
    // other code whit udp_pack_t
}
// other code whit udp_pack_t
getting_not_ip_address:
setting_not_ip_address:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.10 `get_ip_address_source_udp_pack()`

```
char * get_ip_address_source_udp_pack (
    udp_pack_t pack)
```

Function for getting ip address source.

Note

You must call `init_udp_pack` before this.

You can call `set_ip_address_source_udp_pack` before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
---------	-------------	-----------------------

Returns

String or NULL pointer is fail. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
ret = set_ip_address_source_udp_pack(pack, "127.0.0.1");
if (ret == -1)
    goto setting_not_ip_address;
{
    char * ip_address = NULL;
    ip_address = get_ip_address_source_udp_pack(pack);
    if (ip_source == NULL)
        getting_not_ip_address;
    // other code whit udp_pack_t
}
// other code whit udp_pack_t
getting_not_ip_address:
setting_not_ip_address:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.11 get_mac_address_destantion_udp_pack()

```
char * get_mac_address_destantion_udp_pack (
    udp_pack_t pack)
```

Function setting mac address for destantion.

Note

You must call [init_udp_pack](#) before this.

You can call [set_mac_address_destantion_udp_pack](#) before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
---------	-------------	-----------------------

Returns

String or NULL pointer is fail. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
ret = set_mac_address_destantion_udp_pack(pack, "ff:ff:ff:ff:ff:ff");
if (ret == -1)
    goto setting_not_mac_address;
{
    char * mac_address = NULL;
    mac_address = get_mac_address_destantion_udp_pack(udp_pack pack);
    // other code whit udp_pack_t
}
// other code whit udp_pack_t
setting_not_mac_address:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.12 `get_mac_address_source_udp_pack()`

```
char * get_mac_address_source_udp_pack (
    udp_pack_t pack)
```

Function getting mac address for source.

Note

You must call `init_udp_pack` before this.

You can call `set_mac_address_source_udp_pack` before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
---------	-------------	-----------------------

Returns

String or NULL pointer is fail. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
ret = set_mac_address_source_udp_pack(pack, "ff:ff:ff:ff:ff:ff");
if (ret == -1)
    goto setting_not_mac_address;
{
    char * mac_address = NULL;
    mac_address = get_mac_address_source_udp_pack(pack);
    // other code whit udp_pack_t
}
// other code whit udp_pack_t
setting_not_mac_address:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.13 `get_pack_udp_pack()`

```
void * get_pack_udp_pack (
    udp_pack_t pack) [static]
```

Function raw get pointer on UDP package.

Parameters

in, out	<i>pack</i>	UDP package for work.
---------	-------------	-----------------------

Returns

Getting UDP package.

Note

This function is private. Not used outside `udp_lib/udp.c`

5.1.4.14 `get_port_destantion_udp_pack()`

```
char * get_port_destantion_udp_pack (
    udp_pack_t pack)
```

Function for getting port destantion.

Note

You must call `init_udp_pack` before this.

You can call `set_port_destantion_udp_pack` before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
---------	-------------	-----------------------

Returns

String or NULL pointer is fail. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
ret = set_port_destantion_udp_pack(pack, "8001");
if (ret == -1)
    goto setting_not_port_destantion;
{
    char * port_destantion = NULL;
    port_destantion = get_port_destantion_udp_pack(pack);
    if (port_destantion == NULL)
        getting_not_port_destantion;
    // other code whit udp_pack_t
}
// other code whit udp_pack_t
getting_not_port_destantion:
setting_not_port_destantion:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.15 `get_port_source_udp_pack()`

```
char * get_port_source_udp_pack (
    udp_pack_t pack)
```

Function for getting port source.

Note

You must call `init_udp_pack` before this.

You can call `set_port_source_udp_pack` before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
---------	-------------	-----------------------

Returns

String or NULL pointer is fail. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
ret = set_port_source_udp_pack(pack, "8003");
if (ret == -1)
    goto setting_not_port_source;
{
    char * port_source = NULL;
    port_source = get_port_source_udp_pack(pack);
    if (port_source == NULL)
        getting_not_port_source;
    // other code whit udp_pack_t
}
// other code whit udp_pack_t
getting_not_port_source:
setting_not_port_source:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.16 get_size_data_udp_pack()

```
uint16_t get_size_data_udp_pack (
    udp_pack_t pack)
```

Function for getting size data.

Note

You must call [init_udp_pack](#) before this.

You can call [set_data_udp_pack](#) before this.

You can call [add_data_udp_pack](#) before this.

You can call [add_byte_udp_pack](#) before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
---------	-------------	-----------------------

Returns

Length data in UDP package. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
{
    uint16_t size_data = get_size_data_udp_pack(pack);
    // other code whit udp_pack_t
}
// other code whit udp_pack_t
getting_not_size_data:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.17 inet_mac()

```
void * inet_mac (
    const char *const mac_address) [static]
```

Function return mac address in big endian.

Parameters

in	<i>mac_address</i>	String mac address.
----	--------------------	---------------------

Returns

Getting buffer mac address.

Note

This function is private. Not used outside [udp_lib/udp.c](#)

5.1.4.18 inet_port()

```
uint16_t inet_port (
    const char *const port) [static]
```

Function support for parsing port.

Parameters

in	<i>port</i>	Port in string format.
----	-------------	------------------------

Returns

Port for big indian architect.

Note

This function is private. Not used outside [udp_lib/udp.c](#)

5.1.4.19 init_udp_pack()

```
udp_pack_t init_udp_pack (
    void )
```

Function for create object UDP package.

Note

You must call [destroy_udp_pack](#) after this.

Returns

pointer or null on error. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
// other code whit using udp_pack_t
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.20 print_udp_pack()

```
ssize_t print_udp_pack (  
    udp_pack_t pack)
```

Function print all info about udp pack.

Note

You must call [init_udp_pack](#) before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
---------	-------------	-----------------------

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;  
udp_pack_t pack = init_udp_pack();  
if (pack == NULL) {  
    ret = -1;  
    goto get_not_udp_pack;  
}  
ret = print_udp_pack(pack);  
if (ret)  
    goto print_not_udp_pack;  
// other code whit udp_pack_t  
print_not_udp_pack:  
get_not_udp_pack:  
destroy_udp_pack(pack);
```

5.1.4.21 send_udp_pack()

```
ssize_t send_udp_pack (  
    udp_pack_t pack)
```

Function to send UDP package.

Note

You must call [init_udp_pack](#) before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
---------	-------------	-----------------------

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;  
udp_pack_t pack = init_udp_pack();  
if (pack == NULL) {  
    ret = -1;  
    goto get_not_udp_pack;  
}  
send_udp_pack(pack);  
get_not_udp_pack:  
destroy_udp_pack(pack);
```

5.1.4.22 set_data_udp_pack()

```
ssize_t set_data_udp_pack (  
    udp_pack_t pack,  
    const void *const data,  
    uint16_t size)
```

Function overriding data in UDP package.

Note

You must call [init_udp_pack](#) before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
in	<i>data</i>	New data in UDP package.
in	<i>size</i>	Size new data for UDP package.

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;  
const char * const message = "Hello, world";  
udp_pack_t pack = init_udp_pack();  
if (pack == NULL) {  
    ret = -1;  
    goto get_not_udp_pack;  
}  
set_data_udp_pack(pack, message, strlen(message));  
get_not_udp_pack:  
destroy_udp_pack(pack);
```

5.1.4.23 set_file_data_udp_pack()

```
ssize_t set_file_data_udp_pack (  
    udp_pack_t pack,  
    const char *const file_name)
```

Function read data from file in UDP package.

Note

You must call [init_udp_pack](#) before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
in	<i>file_name</i>	File name for read in UDP pack.

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;  
udp_pack_t pack = init_udp_pack();  
if (pack == NULL) {  
    ret = -1;  
    goto get_not_udp_pack;  
}  
set_file_data_udp_pack(pack, "Hello.txt");  
get_not_udp_pack:  
destroy_udp_pack(pack);
```

5.1.4.24 set_input_data_udp_pack()

```
ssize_t set_input_data_udp_pack (  
    udp_pack_t pack)
```

Function read data from stdin in UDP package.

Note

You must call [init_udp_pack](#) before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
---------	-------------	-----------------------

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;  
udp_pack_t pack = init_udp_pack();  
if (pack == NULL) {  
    ret = -1;  
    goto get_not_udp_pack;  
}  
set_input_data_udp_pack(pack);  
get_not_udp_pack:  
destroy_udp_pack(pack);
```

5.1.4.25 set_interface_udp_pack()

```
ssize_t set_interface_udp_pack (  
    udp_pack_t pack,  
    const char *const interface)
```

Function for pick interface to send UDP package.

Note

You must call [init_udp_pack](#) before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
in	<i>interface</i>	Interface to send UDP pack.

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;  
udp_pack_t pack = init_udp_pack();  
if (pack == NULL) {  
    ret = -1;  
    goto get_not_udp_pack;  
}  
set_interface_udp_pack(pack, "lo");  
get_not_udp_pack:  
destroy_udp_pack(pack);
```

5.1.4.26 set_ip_address_destantion_udp_pack()

```
ssize_t set_ip_address_destantion_udp_pack (
    udp_pack_t pack,
    const char *const ip)
```

Function for setting destantion ip address in UDP package.

Note

You must call `init_udp_pack` before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
in	<i>ip</i>	Destination ip address to send.

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
ret = set_ip_address_destantion_udp_pack(pack, "127.0.0.1");
if (ret == -1)
    goto setting_not_ip_address;
// other code whit udp_pack_t
setting_not_ip_address:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.27 set_ip_address_source_udp_pack()

```
ssize_t set_ip_address_source_udp_pack (
    udp_pack_t pack,
    const char *const ip)
```

Function for setting source ip address in UDP package.

Note

You must call `init_udp_pack` before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
in	<i>ip</i>	Source ip to send.

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
ret = set_ip_address_source_udp_pack(pack, "127.0.0.1");
if (ret == -1)
    goto set_not_ip_address;
// other code whit udp_pack_t
set_not_ip_address:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.28 set_mac_address_destantion_udp_pack()

```
ssize_t set_mac_address_destantion_udp_pack (
    udp_pack_t pack,
    const char *const mac_address)
```

Function setting mac address for destantion.

Note

You must call [init_udp_pack](#) before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
in	<i>mac_address</i>	String is mac address destantion.

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
ret = set_mac_address_destantion_udp_pack(pack, "ff:ff:ff:ff:ff:ff");
if (ret == -1)
    goto setting_not_mac_address;
// other code whit udp_pack_t
setting_not_mac_address:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.29 set_mac_address_source_udp_pack()

```
ssize_t set_mac_address_source_udp_pack (
    udp_pack_t pack,
    const char *const mac_address)
```

Function setting mac address for source.

Note

You must call `init_udp_pack` before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
in	<i>mac_address</i>	String is mac address source.

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
ret = set_mac_address_source_udp_pack(pack, "ff:ff:ff:ff:ff:ff");
if (ret == -1)
    goto setting_not_mac_address;
// other code whit udp_pack_t
setting_not_mac_address:
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.30 set_port_destantion_udp_pack()

```
ssize_t set_port_destantion_udp_pack (
    udp_pack_t pack,
    const char *const port)
```

Function for setting destantion port in UDP package.

Note

You must call `init_udp_pack` before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
in	<i>port</i>	Destination port to send.

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;
udp_pack_t pack = init_udp_pack();
if (pack == NULL) {
    ret = -1;
    goto get_not_udp_pack;
}
set_port_destantion_udp_pack(pack, "8003");
get_not_udp_pack:
destroy_udp_pack(pack);
```

5.1.4.31 set_port_source_udp_pack()

```
ssize_t set_port_source_udp_pack (  
    udp_pack_t pack,  
    const char *const port)
```

Function for setting source port in UDP package.

Note

You must call [init_udp_pack](#) before this.

Parameters

in, out	<i>pack</i>	UDP package for work.
in	<i>port</i>	Source port to send.

Returns

0 or -1 on error. Usage example.

```
ssize_t ret = 0;  
udp_pack_t pack = init_udp_pack();  
if (pack == NULL) {  
    ret = -1;  
    goto get_not_udp_pack;  
}  
set_port_source_udp_pack(pack, "8001");  
get_not_udp_pack:  
destroy_udp_pack(pack);
```

5.1.4.32 set_size_udp_pack()

```
void set_size_udp_pack (  
    udp_pack_t pack,  
    const uint16_t size) [static]
```

Function raw write size UDP package.

Parameters

in, out	<i>pack</i>	UDP package for work.
in	<i>size</i>	New size UDP package.

Note

This function is private. Not used outside [udp_lib/udp.c](#)

5.1.4.33 sum_compute()

```
uint32_t sum_compute (
    void * ptr,
    uint16_t nbytes) [static]
```

Function calculate sum in big endian.

Parameters

in	<i>ptr</i>	Buffer for calculating sum in big endian.
in	<i>nbytes</i>	Size buffer for calculating sum.

Returns

Sum buffer in big endian.

Note

This function is private. Not used outside [udp_lib/udp.c](#)

5.1.5 Variable Documentation

5.1.5.1 PACKED

```
struct pseudo\_header PACKED
```


Chapter 6

Class Documentation

6.1 pseudo_header Struct Reference

Struct [pseudo_header](#) for calculate checksum for UDP package.

Public Attributes

- uint32_t [source_address](#)
- uint32_t [dest_address](#)
- uint8_t [placeholder](#)
- uint8_t [protocol](#)
- uint16_t [udp_length](#)

6.1.1 Detailed Description

Struct [pseudo_header](#) for calculate checksum for UDP package.

Note

This struct is private. Not used outside [udp_lib/udp.c](#)

6.1.2 Member Data Documentation

6.1.2.1 dest_address

```
uint32_t pseudo_header::dest_address
```

Destination ip addresses

6.1.2.2 placeholder

```
uint8_t pseudo_header::placeholder
```

Placeholder is 0x00.

6.1.2.3 protocol

```
uint8_t pseudo_header::protocol
```

Constant protocol.

6.1.2.4 source_address

```
uint32_t pseudo_header::source_address
```

Source ip addresses

6.1.2.5 udp_length

```
uint16_t pseudo_header::udp_length
```

Length UDP package.

The documentation for this struct was generated from the following file:

- [udp.c](#)

6.2 udp_head Struct Reference

Struct is header UDP pack.

Public Attributes

- uint16_t [m_port_source](#)
- uint16_t [m_port_destantion](#)
- uint16_t [m_length](#)
- uint16_t [m_checksum](#)

6.2.1 Detailed Description

Struct is header UDP pack.

Note

This struct is private. Not used outside [udp_lib/udp.c](#)

6.2.2 Member Data Documentation

6.2.2.1 m_checksum

```
uint16_t udp_head::m_checksum
```

Calculated software checksum

6.2.2.2 m_length

```
uint16_t udp_head::m_length
```

Length udp pack

6.2.2.3 m_port_destantion

```
uint16_t udp_head::m_port_destantion
```

Port destantion

6.2.2.4 m_port_source

```
uint16_t udp_head::m_port_source
```

Port source

The documentation for this struct was generated from the following file:

- [udp.c](#)

6.3 udp_pack Struct Reference

Struct is UDP package.

Public Attributes

- char [m_interface](#) [IFNAMSIZ]
- struct ethhdr [m_ethhdr](#)
- struct iphdr [m_iphdr](#)
- struct [udp_head](#) [m_head](#)
- uint8_t [m_data](#) [MAX_SIZE_DATA]

6.3.1 Detailed Description

Struct is UDP package.

Note

This struct is private. Not used outside [udp_lib/udp.c](#)

6.3.2 Member Data Documentation

6.3.2.1 m_data

```
uint8_t udp_pack::m_data[MAX_SIZE_DATA]
```

Data in UDP package.

6.3.2.2 m_ethhdr

```
struct ethhdr udp_pack::m_ethhdr
```

Ethernet header start UDP package.

6.3.2.3 m_head

```
struct udp\_head udp_pack::m_head
```

UDP header

6.3.2.4 m_interface

```
char udp_pack::m_interface[IFNAMSIZ]
```

Name ethernet interface.

6.3.2.5 m_iphdr

```
struct iphdr udp_pack::m_iphdr
```

IP header

The documentation for this struct was generated from the following file:

- [udp.c](#)

Chapter 7

File Documentation

7.1 README.md File Reference

7.2 udp.c File Reference

Code file for work udp package.

```
#include "udp.h"
#include <stdint.h>
#include <string.h>
#include <stdlib.h>
#include <stdio.h>
#include <sys/ioctl.h>
#include <sys/stat.h>
#include <unistd.h>
#include <fcntl.h>
#include <errno.h>
#include <stddef.h>
#include <net/if.h>
#include <linux/if_packet.h>
#include <net/ethernet.h>
#include <arpa/inet.h>
#include <sys/socket.h>
#include <netinet/ip.h>
```

Classes

- struct [udp_head](#)
Struct is header UDP pack.
- struct [udp_pack](#)
Struct is UDP package.
- struct [pseudo_header](#)
Struct [pseudo_header](#) for calculate checksum for UDP package.

Macros

- #define `PACKED` `__attribute__((packed))`
- #define `HEAD_ETH` `sizeof(struct ethhdr)`
- #define `HEAD_UDP` `sizeof(struct udp_head)`
- #define `HEAD_IP` `sizeof(struct iphdr)`
- #define `HEAD_UDP_IP` `(HEAD_IP + HEAD_UDP)`
- #define `MAX_SIZE_DATA` `(0xFFFF - HEAD_UDP_IP)`
- #define `NULL_CHECKSUM` `0x0000`
- #define `MIN`(left, right)
- #define `HEAD_PSEUDO` `sizeof(struct pseudo_header)`

Functions

- `udp_pack_t` `init_udp_pack` (void)
Function for create object UDP package.
- static uint16_t `inet_port` (const char *const port)
Function support for parsing port.
- ssize_t `set_port_source_udp_pack` (`udp_pack_t` pack, const char *const port)
Function for setting source port in UDP package.
- ssize_t `set_port_destantion_udp_pack` (`udp_pack_t` pack, const char *const port)
Function for setting destantion port in UDP package.
- ssize_t `set_ip_address_source_udp_pack` (`udp_pack_t` pack, const char *const ip)
Function for setting source ip address in UDP package.
- ssize_t `set_ip_address_destantion_udp_pack` (`udp_pack_t` pack, const char *const ip)
Function for setting destantion ip address in UDP package.
- static void `set_size_udp_pack` (`udp_pack_t` pack, const uint16_t size)
Function raw write size UDP package.
- uint16_t `get_size_data_udp_pack` (`udp_pack_t` pack)
Function for getting size data.
- ssize_t `add_data_udp_pack` (`udp_pack_t` pack, void *data, const uint16_t size)
Function addition data in UDP package.
- ssize_t `set_data_udp_pack` (`udp_pack_t` pack, const void *const data, uint16_t size)
Function overriding data in UDP package.
- ssize_t `add_byte_udp_pack` (`udp_pack_t` pack, uint8_t byte)
Function addition byte in UDP package.
- ssize_t `set_input_data_udp_pack` (`udp_pack_t` pack)
Function read data from stdin in UDP package.
- ssize_t `set_file_data_udp_pack` (`udp_pack_t` pack, const char *const file_name)
Function read data from file in UDP package.
- static uint32_t `sum_compute` (void *ptr, uint16_t nbytes)
Function calculate sum in big endian.
- static uint16_t `checksum_compute` (uint32_t sum)
Function calculating from sum big endian to checksum big endian.
- static void `calculate_checksum_udp_pack` (`udp_pack_t` pack)
Function calculate checksum for ip header and UDP package.
- ssize_t `set_interface_udp_pack` (`udp_pack_t` pack, const char *const interface)
Function for pick interface to send UDP package.
- static void * `get_pack_udp_pack` (`udp_pack_t` pack)
Function raw get pointer on UDP package.
- ssize_t `send_udp_pack` (`udp_pack_t` pack)

- Function to send UDP package.*

 - static void * [inet_mac](#) (const char *const mac_address)

Function return mac address in big endian.
- ssize_t [set_mac_address_source_udp_pack](#) (udp_pack_t pack, const char *const mac_address)

Function setting mac address for source.
- ssize_t [set_mac_address_destantion_udp_pack](#) (udp_pack_t pack, const char *const mac_address)

Function setting mac address for destantion.
- char * [get_ip_address_source_udp_pack](#) (udp_pack_t pack)

Function for getting ip address source.
- char * [get_ip_address_destantion_udp_pack](#) (udp_pack_t pack)

Function for getting ip address destantion.
- char * [get_interface_udp_pack](#) (udp_pack_t pack)

Function for getting interface.
- char * [get_mac_address_source_udp_pack](#) (udp_pack_t pack)

Function getting mac address for source.
- char * [get_mac_address_destantion_udp_pack](#) (udp_pack_t pack)

Function setting mac address for destantion.
- char * [get_port_destantion_udp_pack](#) (udp_pack_t pack)

Function for getting port destantion.
- char * [get_port_source_udp_pack](#) (udp_pack_t pack)

Function for getting port source.
- ssize_t [print_udp_pack](#) (udp_pack_t pack)

Function print all info about udp pack.
- char * [get_data_udp_pack](#) (udp_pack_t pack)

Function for getting data.
- void [destroy_udp_pack](#) (udp_pack_t pack)

Function free UDP package.

Variables

- struct [udp_head](#) PACKED

7.2.1 Detailed Description

Code file for work udp package.

Author

Vladsanin777

7.2.2 Macro Definition Documentation

7.2.2.1 HEAD_ETH

```
#define HEAD_ETH sizeof(struct ethhdr)
```

7.2.2.2 HEAD_IP

```
#define HEAD_IP sizeof(struct iphdr)
```

7.2.2.3 HEAD_PSEUDO

```
#define HEAD_PSEUDO sizeof(struct pseudo_header)
```

7.2.2.4 HEAD_UDP

```
#define HEAD_UDP sizeof(struct udp_head)
```

7.2.2.5 HEAD_UDP_IP

```
#define HEAD_UDP_IP (HEAD_IP + HEAD_UDP)
```

7.2.2.6 MAX_SIZE_DATA

```
#define MAX_SIZE_DATA (0xFFFF - HEAD_UDP_IP)
```

7.2.2.7 MIN

```
#define MIN(  
    left,  
    righth)
```

Value:

```
((left) < ((typeof(left))righth)) ? (left) : ((typeof(left))righth)
```

7.2.2.8 NULL_CHECKSUM

```
#define NULL_CHECKSUM 0x0000
```

7.3 udp.h File Reference

Header file for work udp package.

```
#include <stdint.h>  
#include <stdlib.h>  
#include <stddef.h>
```

Typedefs

- typedef struct [udp_pack](#) * [udp_pack_t](#)
UDP packet descriptor.

Functions

- [udp_pack_t](#) [init_udp_pack](#) (void)
Function for create object UDP package.
- ssize_t [set_port_source_udp_pack](#) ([udp_pack_t](#) pack, const char *const port)
Function for setting source port in UDP package.
- ssize_t [set_port_destantion_udp_pack](#) ([udp_pack_t](#) pack, const char *const port)
Function for setting destantion port in UDP package.
- ssize_t [set_ip_address_source_udp_pack](#) ([udp_pack_t](#) pack, const char *const ip)
Function for setting source ip address in UDP package.
- ssize_t [set_ip_address_destantion_udp_pack](#) ([udp_pack_t](#) pack, const char *const ip)
Function for setting destantion ip address in UDP package.
- ssize_t [add_data_udp_pack](#) ([udp_pack_t](#) pack, void *data, uint16_t size)
Function addition data in UDP package.
- ssize_t [set_data_udp_pack](#) ([udp_pack_t](#) pack, const void *const data, uint16_t size)
Function overriding data in UDP package.
- ssize_t [add_byte_udp_pack](#) ([udp_pack_t](#) pack, const uint8_t byte)
Function addition byte in UDP package.
- ssize_t [set_input_data_udp_pack](#) ([udp_pack_t](#) pack)
Function read data from stdin in UDP package.
- ssize_t [set_file_data_udp_pack](#) ([udp_pack_t](#) pack, const char *const file_name)
Function read data from file in UDP package.
- ssize_t [set_interface_udp_pack](#) ([udp_pack_t](#) pack, const char *const interface)
Function for pick interface to send UDP package.
- ssize_t [send_udp_pack](#) ([udp_pack_t](#) pack)
Function to send UDP package.
- char * [get_mac_address_source_udp_pack](#) ([udp_pack_t](#) pack)
Function getting mac address for source.
- char * [get_mac_address_destantion_udp_pack](#) ([udp_pack_t](#) pack)
Function setting mac address for destantion.
- ssize_t [set_mac_address_source_udp_pack](#) ([udp_pack_t](#) pack, const char *const mac_address)
Function setting mac address for source.
- ssize_t [set_mac_address_destantion_udp_pack](#) ([udp_pack_t](#) pack, const char *const mac_address)
Function setting mac address for destantion.
- char * [get_interface_udp_pack](#) ([udp_pack_t](#) pack)
Function for getting interface.
- char * [get_port_source_udp_pack](#) ([udp_pack_t](#) pack)
Function for getting port source.
- char * [get_port_destantion_udp_pack](#) ([udp_pack_t](#) pack)
Function for getting port destantion.
- char * [get_ip_address_source_udp_pack](#) ([udp_pack_t](#) pack)
Function for getting ip address source.
- char * [get_ip_address_destantion_udp_pack](#) ([udp_pack_t](#) pack)
Function for getting ip address destantion.
- uint16_t [get_size_data_udp_pack](#) ([udp_pack_t](#) pack)

Function for getting size data.

- char * [get_data_udp_pack](#) (udp_pack_t pack)

Function for getting data.

- char * [get_data_hex_udp_pack](#) (udp_pack_t pack)

Function for getting data hex.

- ssize_t [print_udp_pack](#) (udp_pack_t pack)

Function print all info about udp pack.

- void [destroy_udp_pack](#) (udp_pack_t pack)

Function free UDP package.

7.3.1 Detailed Description

Header file for work udp package.

Author

Vladsanin777

7.4 udp.h

[Go to the documentation of this file.](#)

```

00001
00006
00007 #ifndef __UDP__H__
00008 #define __UDP__H__
00009
00010 #include <stdint.h>
00011 #include <stdlib.h>
00012 #include <stddef.h>
00013
00014 #ifdef __cplusplus
00015
00016 extern "C" {
00017
00018 #endif
00019
00025
00029 struct udp_pack;
00030
00036 typedef struct udp_pack * udp_pack_t;
00037
00055 udp_pack_t init_udp_pack(void);
00056
00076 ssize_t set_port_source_udp_pack(udp_pack_t pack, const char * const port);
00077
00097 ssize_t set_port_destantion_udp_pack(udp_pack_t pack, const char * const port);
00098
00122 ssize_t set_ip_address_source_udp_pack( \
00123     udp_pack_t pack, const char * const ip);
00124
00148 ssize_t set_ip_address_destantion_udp_pack( \
00149     udp_pack_t pack, const char * const ip);
00150
00175 ssize_t add_data_udp_pack(udp_pack_t pack, void * data, uint16_t size);
00176
00198 ssize_t set_data_udp_pack(udp_pack_t pack, const void * const data, \
00199     uint16_t size);
00200
00225 ssize_t add_byte_udp_pack(udp_pack_t pack, \
00226     const uint8_t byte);
00227
00246 ssize_t set_input_data_udp_pack(udp_pack_t pack);
00247
00267 ssize_t set_file_data_udp_pack( \
00268     udp_pack_t pack, const char * const file_name);
00269
00289 ssize_t set_interface_udp_pack( \
00290     udp_pack_t pack, const char * const interface);

```

```
00291
00310 ssize_t send_udp_pack(udp_pack_t pack);
00311
00340 char * get_mac_address_source_udp_pack(udp_pack_t pack);
00341
00370 char * get_mac_address_destantion_udp_pack(udp_pack_t pack);
00371
00395 ssize_t set_mac_address_source_udp_pack(udp_pack_t pack, \
00396     const char * const mac_address);
00397
00421 ssize_t set_mac_address_destantion_udp_pack(udp_pack_t pack, \
00422     const char * const mac_address);
00423
00455 char * get_interface_udp_pack(udp_pack_t pack);
00456
00488 char * get_port_source_udp_pack(udp_pack_t pack);
00489
00521 char * get_port_destantion_udp_pack(udp_pack_t pack);
00522
00554 char * get_ip_address_source_udp_pack(udp_pack_t pack);
00555
00587 char * get_ip_address_destantion_udp_pack(udp_pack_t pack);
00588
00615 uint16_t get_size_data_udp_pack(udp_pack_t pack);
00616
00639 char * get_data_udp_pack(udp_pack_t pack);
00640
00668 char * get_data_hex_udp_pack(udp_pack_t pack);
00669
00692 ssize_t print_udp_pack(udp_pack_t pack);
00693
00711 void destroy_udp_pack(udp_pack_t pack);
00712
00714
00715 #ifdef __cplusplus
00716
00717 }
00718
00719 #endif
00720
00721 #endif
```


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